
Exercises: Advanced Lesson 7

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Your Math Class: MATH 7013 – Advanced Mathematical Modeling

1: Determine the Head and FullForm of the expressions: e^x and x^e

```
Clear[x];  
{Head[ex], FullForm[ex]}  
{Head[xe], FullForm[xe]}
```

2: Determine the Head FullForm and TreeForm of the expression: $x + y^2 \{u, v, w\}$ then evaluate the expression.

```
Clear[x, y, u, v, w];  
x + y2 {u, v, w}  
Head[x + y2 {u, v, w}]  
FullForm[x + y2 {u, v, w}]  
TreeForm[x + y2 {u, v, w}]
```

3: Use the Head command to determine if π is stored internally as an integer, a rational number, an approximate real number, etc. Using a Boolean function, determine if *Mathematica's* internal representation of π is designated as a numeric type. Using a Boolean function, determine if π is positive.

```
Head[ $\pi$ ]  
NumericQ[ $\pi$ ]  
NumberQ[ $\pi$ ]  
Positive[ $\pi$ ]
```

4: Using Boolean functions, determine if "This is a string" is a list or is a string.

```
ListQ["This is a string"]  
StringQ["This is a string"]
```

5: Using Boolean functions, determine if 3 is a member of {1, 2, 3, 4, 5}.

```
MemberQ[{1, 2, 3, 4, 5}, 3]
```

6: Using Boolean functions, define a function "f" such that:

$f(x) = +1$ when x is positive,
 $f(0) = 0$ (do not use a Boolean function here),

$f(x) = -1$ when x is negative.

Use ?? to determine the assignments made to "f" (note the order from top down).

Evaluate: $f[-2]$, $f[0.023]$, $f[0]$, $f[0.0]$, $f[\pi]$, $f[3 + 4 \cdot I]$, $f[a]$

Plot the function $f(x)$ for $-2 \leq x \leq +2$

```
Clear[f, x];
f[x_] := +1 /; Positive[x];
f[0] = 0;
f[x_] := -1 /; Negative[x]

??f

{f[-2], f[0.023`], f[0], f[0.0`], f[π], f[3 + 4 I], f[a]}

Plot[f[x], {x, -2, 2}]
```

7: Check the Attributes of the Plot and Integrate commands.

Attempt to evaluate:

```
x = 2
Plot[x^2, {x, 0, 1}]
Plot[∫ Sin[x] dx, {x, 0, 1}]
```

Why did the first Plot command work?

What must be done and why must it be done so that the second Plot command will work?

Why did the assignment $x = 2$ have no effect on the Plot command, but did have an effect on the Integrate command?

```
Attributes[Plot]
```

```
Attributes[Integrate]
```

```
Clear[x];
x = 2;
Plot[x^2, {x, 0, 1}]

Plot[∫ Sin[x] dx, {x, 0, 1}]

Clear[x];
Plot[Evaluate[∫ Sin[x] dx], {x, 0, 1}]
```

Plot holds its first argument and uses its own local x , so the global $x = 2$ never reaches it. That is why the first Plot works. Integrate does not hold its arguments, so inside the second Plot the integral is handed to Integrate with x already equal to 2, making it an integral with respect to the number 2, which fails at every sample point. The fix is to force the integral to be worked out first, with $\text{Plot}[\text{Evaluate}[\dots]]$, after clearing x . Note that in this version $\text{Attributes}[\text{Plot}]$ reports only Protected and ReadProtected, but the holding behaviour is still there.

8: Part 1: Check the Attributes of the Table and Integrate commands.

Part 2: Attempt to evaluate:

```
x = 2
n = 3
Table[ $\int \sin[n x] dx$ , {n, 0, 10}]
```

Part 3: Make this code work!

Part 4: Explain at what point in the construction of the table were the integrals evaluated, and how this relates to the attributes of the table command.

Part 5: Explain how the integrals were evaluated (when $x = 2$) and how this relates to the attributes of the integrate command.

Part 6: Why did the assignment $x = 2$ have no effect on the Table command but did have an effect on the Integrate command?

Part 7: Why did the assignment $n = 3$ have no effect on the Table command?

```
Attributes[Table]
```

```
Attributes[Integrate]
```

```
Clear[x, n];
x = 2;
n = 3;
Table[ $\int \sin[n x] dx$ , {n, 0, 10}]

Clear[x];
n = 3;
Table[ $\int \sin[n x] dx$ , {n, 0, 10}]
```

Table has the attribute HoldAll, so it does not evaluate its body in advance: it substitutes each n and evaluates then. Integrate holds nothing, so its arguments are evaluated before it sees them. With $x = 2$ the integration variable became the number 2 and every integral in the table was left unevaluated. Clearing x makes the table work. The assignment $n = 3$ caused no trouble, because Table holds its body and supplies its own n from the iterator.

9: Part 1: Evaluate: $x + y^2 \{u, v\}$ and observe the final form.

Observe that it is different than the initial form.

Part 2: Evaluate: **TreeForm**[$x + y^2 \{u, v\}$] and reconcile this expression with the first part.

Part 3: Locate the relative position of $x + u y^2$ in $x + y^2 \{u, v\}$

Part 4: Locate the relative position of $u y^2$ in $x + y^2 \{u, v\}$

Part 5: Locate the relative position of y^2 in $x + y^2 \{u, v\}$

Part 6: Locate the relative position of u in $x + y^2 \{u, v\}$

Part 7: Locate the relative position of y in $x + y^2 \{u, v\}$

Part 8: Evaluate the following code and relate what you see to the previous parts:

```
Clear[x, y, u, v]
Table[( $x + y^2 \{u, v\}$ )[[i]] // TreeForm, {i, 1, 2}] // TableForm
Table[( $x + y^2 \{u, v\}$ )[[i, j]] // TreeForm, {i, 1, 2}, {j, 1, 2}] // TableForm
```

Part 9: Evaluate the following code one line at a time and relate what you see to the previous parts:

```
{x + u y2, x + v y2} // TreeForm
{x + u y2, x + v y2} [[0]]
{x + u y2, x + v y2} [[1, 0]]
{x + u y2, x + v y2} [[1, 2, 0]]
{x + u y2, x + v y2} [[1, 2, 2, 0]]
```

```
Clear[x, y, u, v];
```

```
x + y2 {u, v}
```

```
TreeForm[x + y2 {u, v}]
```

```
Position[x + y2 {u, v}, x + u y2]
```

```
Position[x + y2 {u, v}, u y2]
```

```
Position[x + y2 {u, v}, y2]
```

```
Position[x + y2 {u, v}, u]
```

```
Position[x + y2 {u, v}, y]
```

```
Clear[x, y, u, v];
```

```
TableForm[Table[TreeForm[(x + y2 {u, v})[[i]], {i, 1, 2}]]
```

```
TableForm[Table[TreeForm[(x + y2 {u, v})[[i, j]], {i, 1, 2}, {j, 1, 2}]]
```

```
Clear[x, y, u, v];
```

```
TreeForm[{x + u y2, x + v y2}]
```

```
{x + u y2, x + v y2}[[0]]
```

```
{x + u y2, x + v y2}[[1, 0]]
```

```
{x + u y2, x + v y2}[[1, 2, 0]]
```

```
{x + u y2, x + v y2}[[1, 2, 2, 0]]
```